MONAURAL CONTINUOUS-RADIUS REGIONAL SPEECH EXTRACTION WITH CROSS-RADIUS CONSISTENCY LEARNING

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ABSTRACT

Source-to-microphone distance enables speaker-independent speech extraction without prior enrollment. We propose a monaural continuous-radius regional speech extraction method that directly models the cumulative speech target within a queried radius. To realize continuous region control, the query radius is encoded as a continuous scalar and injected into a time-frequency extraction network, allowing a single model to operate over arbitrary radii within the trained range. Exploiting the nested structure of target-speaker sets across query radii, we introduce cross-radius consistency learning to stabilize predictions for adjacent radii sharing the same nonempty target-speaker set. Experiments on measured RIRs show that continuous-radius conditioning improves selective extraction over discrete conditioning. The proposed method achieves 29.25 dB SI-SDR, 4.40 dB SI-SDRi, 62.01 dB attenuation, and 0.46% RCE, outperforming fixed-threshold and local-range baselines. It also remains effective with more speakers and additive noise.

***Index Terms*—** Regional speech extraction, radius conditioning, continuous radius, consistency learning

1. INTRODUCTION

Target speech extraction (TSE) recovers speech of interest from an acoustic mixture according to a user-specified cue. TSE is essential for speech communication and listening applications in multi-talker environments. Conventional TSE often relies on speaker identity [1, 2], requiring prior enrollment or target-specific information and limiting flexibility when the desired speaker is unknown or changes dynamically. Spatial cues provide an alternative by specifying target speech according to location rather than identity. In particular, source-to-microphone distance selects speech by proximity to the recording device without relying on speaker identity [3, 4].

Recent studies have shown that source-to-microphone distance enables monaural speech extraction. Patterson et al. [5] introduced monaural distance-based sound separation (DSS), which separates near and far sources relative to a fixed distance threshold. Lin et al. [6] incorporated distance and speaker information into monaural TSE. More recently, Shi et al. [7, 8] developed query-distance-based single-channel target speech extraction (QD-TSE) for local-range extraction, with room-aware conditioning improving robustness. However, existing monaural formulations use fixed thresholds or local distance windows rather than a continuously controlled cumulative region.

Regional speech extraction has also been explored using multichannel spatial information. Gu and Luo [9] proposed ReZero, which extracts sounds from user-specified spatial regions using microphone arrays. Chen et al. [10] developed programmable sound bubbles for hearable devices, preserving sounds within a specified radius while suppressing those outside. More recent multichannel systems combine directional, distance, and reverberation cues for regional extraction [11, 12]. However, these approaches require multichannel observations, leaving continuous-radius monaural regional extraction underexplored.

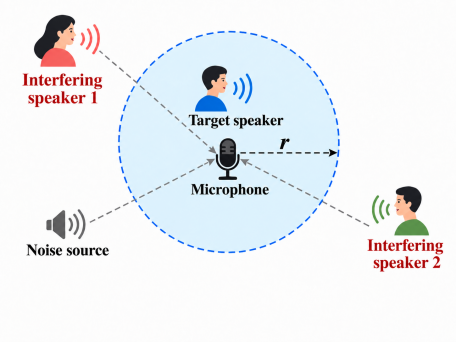


Fig. 1. Continuous-radius regional speech extraction task.

In this work, we investigate monaural continuous-radius regional speech extraction, which preserves speech within a queried region while suppressing speech outside it, as illustrated in Fig. 1. The query radius defines a spatial region around the microphone, and the task is to extract all speakers within this region from the monaural mixture. Unlike fixed-threshold DSS and local-range QD-TSE, our approach directly models cumulative regional targets and supports continuous-radius queries within the trained range using a single model. Demo samples and code are available at `https://dongb001.github.io/CRadius-RSE/`.

The main contributions of this work are threefold:

- We propose a monaural continuous-radius regional speech extraction method that directly models cumulative regional targets with continuous radius conditioning in a single model.
- We exploit the nested structure of target-speaker sets across query radii and introduce cross-radius consistency learning (CRC) to encourage consistent predictions between adjacent radii sharing the same nonempty target-speaker set.
- Experiments on measured RIRs show that the proposed method outperforms fixed-threshold and local-range baselines across the regional extraction tasks, while remaining effective with more speakers and additive noise.

2. METHOD

2.1. Problem formulation

Consider a monaural acoustic scene containing N speakers. The observed mixture is

$$x(t) = \sum_{i=1}^{N} s_i(t) + n(t), \tag{1}$$

where $x(t)$ denotes the monaural mixture, $s_i(t)$ the microphone-received signal from the ith speaker, and $n(t)$ additive noise. Let d_i denote the source-to-microphone distance for speaker i. Given a query radius $r \in [0, R_{\max}]$, where $R_{\max} = 4$ m is the maximum query radius used during training and evaluation, we define the target-speaker set as

$$\mathcal{S}(r) = \{i \mid d_i \le r\}. \tag{2}$$

The desired signal is the sum of speakers within the query radius:

$$y_r(t) = \sum_{i \in \mathcal{S}(r)} s_i(t). \tag{3}$$

When $\mathcal{S}(r) = \varnothing$, the target is silent: $y_r(t) = 0$. The source distances $\{d_i\}_{i=1}^{N}$ are used only to construct supervision during training and evaluation. At inference, the model receives a user-specified physical radius r and does not require source-distance estimation.

Following target speech extraction formulations [1, 2, 6, 7], our objective is to learn a model f_θ that estimates the desired signal from the mixture x and query radius r,

$$\hat{y}_r(t) = f_\theta(x, r), \tag{4}$$

where $\hat{y}_r(t)$ denotes the estimated waveform.

The continuous-radius formulation enables a single model to handle continuous spatial extents within $[0, R_{\max}]$ without restricting queries to a discrete set of radii. For any $0 \le r_1 < r_2 \le R_{\max}$, the target-speaker sets satisfy

$$\mathcal{S}(r_1) \subseteq \mathcal{S}(r_2), \tag{5}$$

inducing a nested structure across radii. This structure has two key implications. First, when the radius increases without crossing any source boundary, the target-speaker set remains unchanged:

$$\mathcal{S}(r_1) = \mathcal{S}(r_2). \tag{6}$$

Consequently, the desired signals coincide:

$$y_{r_1}(t) = y_{r_2}(t). \tag{7}$$

Second, when the radius increment introduces new speakers, the desired signal changes by the sum of the newly included sources. This piecewise-invariant behavior motivates the cross-radius consistency learning introduced in Sec. 2.3.

2.2. Extraction network with continuous-radius conditioning

The proposed continuous-radius regional speech extraction network is illustrated in Fig. 2. We employ the lightweight TF-GridNet [13] backbone. Given a mixture waveform x, we compute its short-time Fourier transform (STFT) $X \in \mathbb{C}^{F \times T}$, where F and T denote frequency bins and time frames. The magnitude-normalized real, imaginary, and log-magnitude components are concatenated

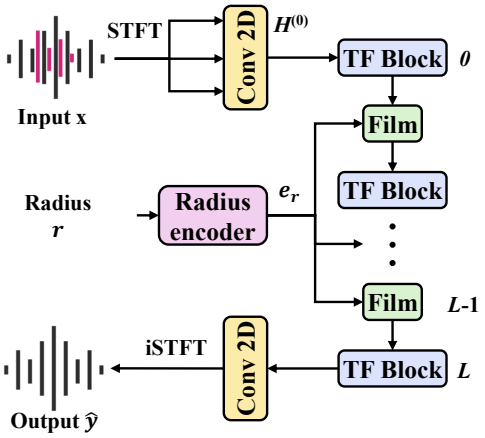


Fig. 2. Continuous-radius extraction network.

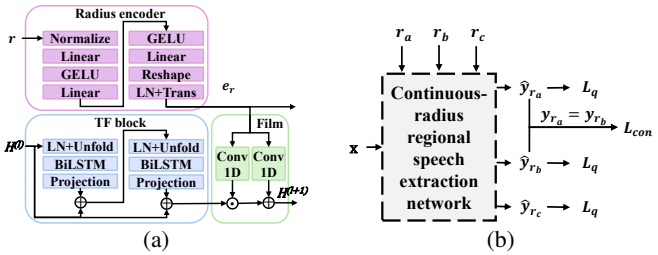


Fig. 3. (a) TF block with radius conditioning and (b) cross-radius consistency learning.

and projected into $H^{(0)} \in \mathbb{R}^{C \times T \times F}$, where C denotes feature channels.

The backbone consists of six sequentially stacked TF blocks indexed by $l = 0, \ldots, L$, where $L = 5$. As shown in Fig. 3(a), given the input $H^{(l)}$ to the lth block, bidirectional sequence modeling is first performed along the frequency axis:

$$U^{(l)} = H^{(l)} + P_l^f\left(\mathrm{BiLSTM}_l^f\left(\mathrm{LN}_l^f(H^{(l)})\right)\right), \quad l = 0, \ldots, L. \tag{8}$$

A second stage is then performed along the temporal axis:

$$V^{(l)} = U^{(l)} + P_l^t\left(\mathrm{BiLSTM}_l^t\left(\mathrm{LN}_l^t(U^{(l)})\right)\right), \quad l = 0, \ldots, L. \tag{9}$$

Here, $\mathrm{LN}(\cdot)$, $\mathrm{BiLSTM}(\cdot)$, and $P(\cdot)$ denote layer normalization, bidirectional LSTM, and linear projection, with superscripts f and t indicating frequency and temporal paths.

The query radius is normalized and mapped by a three-layer radius encoder E_ϕ to a frequency-dependent representation:

$$\tilde{r} = r / R_{\max}, \qquad e_r = E_\phi(\tilde{r}). \tag{10}$$

The representation e_r is injected between consecutive TF blocks through feature-wise linear modulation (FiLM) [14]:

$$H^{(l)} = \Gamma_{l-1}(e_r) \odot V^{(l-1)} + B_{l-1}(e_r), \quad l = 1, \ldots, L, \tag{11}$$

where $H^{(0)}$ is the input to block 0, while $H^{(l)}$ for $l \ge 1$ is the FiLM-conditioned input to block l. The functions $\Gamma_{l-1}(\cdot)$ and $B_{l-1}(\cdot)$ generate scaling and bias parameters, respectively, and $\odot$ denotes element-wise multiplication. Thus, the five FiLM modules condition only the transitions between adjacent TF blocks, with no FiLM

before block 0. The output $V^{(L)}$ of the last TF block is decoded into a complex spectrum and reconstructed as the waveform $\hat{y}_r(t)$ using an inverse STFT.

2.3. Cross-radius consistency learning

To explicitly exploit the nested structure across regions, we sample Q query radii $r_1 < r_2 < \cdots < r_Q$ from the continuous interval $[0, R_{\max}]$ for the same mixture x. In our implementation, we set $Q = 4$ and sample radii uniformly. Following the local invariance established in Sec. 2.1, consistency is imposed only on adjacent radius pairs whose target-speaker sets remain unchanged:

$$\mathcal{P} = \{(r_q, r_{q+1}) \mid \mathcal{S}(r_q) = \mathcal{S}(r_{q+1}) \neq \varnothing,\ 1 \leq q < Q\}. \quad (12)$$

If increasing the radius introduces additional speakers into the region, the desired signal should change accordingly, and the corresponding pair is excluded from the consistency constraint to avoid oversmoothing across speaker boundaries. Computing valid radius pairs $\mathcal{P}$ requires ground-truth source distances during training, available in simulated data where positions are known. We further exclude pairs whose common-target RMS does not exceed $\tau = 10^{-4}$.

As illustrated in Fig. 3(b), for each valid pair $(r_a, r_b) \in \mathcal{P}$, we minimize the mean absolute waveform discrepancy normalized by the RMS amplitude of their common target:

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{P}|} \sum_{(r_a, r_b) \in \mathcal{P}} \frac{\|\hat{y}_{r_a} - \hat{y}_{r_b}\|_1}{N_s \max(\text{RMS}(y_{r_a}), \epsilon)}. \quad (13)$$

Here, N_s is the waveform length and $|\mathcal{P}|$ is the number of valid pairs. We set $\mathcal{L}_{\text{con}} = 0$ when $\mathcal{P} = \varnothing$. In implementation, the RMS normalization term is lower-bounded by $\epsilon = 10^{-8}$ for numerical stability. The root-mean-square amplitude is defined as

$$\text{RMS}(z) = \sqrt{\text{mean}(z^2)}. \quad (14)$$

2.4. Loss function

For each query radius r_q, the loss depends on whether the target region contains any speaker. We denote $\hat{y}_q = \hat{y}_{r_q}$, $y_q = y_{r_q}$, and the prediction error $g_q = \hat{y}_q - y_q$. When the target region is nonempty, we use a normalized reconstruction loss:

$$\ell_{\text{rec}}(\hat{y}_q, y_q) = 10 \log_{10} \left(1 + \frac{\text{MSE}(g_q)}{\text{MSE}(y_q) + \epsilon}\right), \quad (15)$$

where $\text{MSE}(z) = \text{mean}(z^2)$ and $\epsilon = 10^{-8}$. The per-radius loss is then defined as

$$\mathcal{L}_q = \begin{cases} \ell_{\text{rec}}(\hat{y}_q, y_q), & \mathcal{S}(r_q) \neq \varnothing, \\ \alpha \cdot \text{mean}(|\hat{y}_q|), & \mathcal{S}(r_q) = \varnothing, \end{cases} \quad (16)$$

where $\alpha = 100$ is the zero-target suppression weight. For empty targets, this penalty term encourages the model to output silence. The extraction loss is obtained by averaging over the Q query radii:

$$\mathcal{L}_{\text{ext}} = \frac{1}{Q} \sum_{q=1}^{Q} \mathcal{L}_q. \quad (17)$$

The total training loss is

$$\mathcal{L} = \mathcal{L}_{\text{ext}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}, \quad (18)$$

where $\mathcal{L}_{\text{con}}$ is the cross-radius consistency loss from Eq. (13), and $\lambda_{\text{con}} = 0.1$.

3. EXPERIMENTAL SETUP

3.1. Data

We generate RIRs for 1,000 rooms using gpuRIR [15], with room length and width sampled from $[2, 12]$ m, height from $[2, 5]$ m, reverberation time RT60 $\in [0.1, 1.0]$ s, and source distances in $[0.3, 5.0]$ m for training and validation. Measured RIRs from dEchorate [16] are used as the test set to evaluate generalization to real acoustic conditions [17]. Speech sources are drawn from AISHELL-1 [18], AISHELL-2 [19], THCHS-30 [20], and VCTK [21]. Training and validation mixtures contain two speakers. Additive noise is sampled from the DNS Challenge noise dataset [22]. All signals are sampled at 16 kHz and are 4.5 s long. Source distances are grouped into five intervals to construct diverse source profiles, while query radii for training and validation are sampled from $[0, R_{\max}]$. The target is the sum of speakers inside the query radius. Noise is added with probability 0.5 at an SNR sampled from $[0, 15]$ dB.

3.2. Baselines

We adopt two monaural distance-based baselines. For DSS [5], we train four fixed-radius models at 1, 2, 3, and 4 m and use the model corresponding to the requested radius at inference. Its near output is treated as the regional estimate. We additionally reproduce QD-TSE [7], which extracts speakers within a local distance window $[d_q - r_{\text{spk}}, d_q + r_{\text{spk}}]$ centered at query distance d_q. Following the original work, we set $r_{\text{spk}} = 0.5$ m. To approximate our cumulative regional query at radius r, we partition $[0, r]$ into non-overlapping intervals of width $\Delta = 1.0$ m and sum the QD-TSE outputs from each interval center. This accumulation requires multiple forward passes per query, whereas our method processes each continuous-radius query in a single forward pass. All methods are evaluated on the same test mixtures at query radii 1, 2, 3, and 4 m.

3.3. Evaluation metrics

We evaluate three conditions based on the target-speaker set induced by the query radius. For full-target cases with all sources within the radius, we measure speech preservation using SI-SDR [23]. For partial-target cases with only some sources included, we evaluate selective extraction using SI-SDRi. For empty-target cases with no sources in the region, we quantify suppression using attenuation (Atten.). To measure cross-radius consistency, we evaluate Radius Consistency Error (RCE) on a 0.1-m query grid over $[0, R_{\max}]$:

$$\text{RCE}(\%) = \frac{100}{|\mathcal{P}_{\text{eval}}|} \sum_{(r_a, r_b) \in \mathcal{P}_{\text{eval}}} \frac{\|\hat{y}_{r_a} - \hat{y}_{r_b}\|_1}{N_s \max(\text{RMS}(x), \epsilon)}. \quad (19)$$

where $\mathcal{P}_{\text{eval}}$ contains adjacent query pairs with the same nonempty target-speaker set, and N_s is the signal length. We report the median RCE over utterances with $|\mathcal{P}_{\text{eval}}| > 0$.

3.4. Implementation details

The FFT length and hop size of the STFT are set to 192 and 128 samples, respectively. The model uses six TF blocks ($L = 5$), a backbone feature dimension of 32, and a bidirectional LSTM hidden dimension of 64, resulting in approximately 0.66 M trainable parameters. We train the model with the Adam optimizer using an initial learning rate of 6×10^{-4}, a warm-up of 2,000 optimization steps, and a ReduceLROnPlateau scheduler. The batch size is 1, and the maximum number of training epochs is 50. Results are averaged over three independent runs with different random seeds.

4. RESULTS

4.1. Comparison with baselines

We assess whether direct cumulative-region modeling improves extraction over task-matched fixed-threshold and local-range baselines on measured RIRs (Table 1). The proposed method achieves the highest scores on all three regional extraction metrics. Compared with DSS, it improves full-target SI-SDR by 7.22 dB, partial-target SI-SDRi by 4.01 dB, and empty-target attenuation by 5.89 dB. Compared with QD-TSE with interval summation, it yields gains of 3.26 dB SI-SDR, 3.28 dB SI-SDRi, and 7.91 dB attenuation. In addition, the proposed model produces each regional estimate with a single forward pass, whereas the interval-summation baseline requires multiple local-range queries.

Table 1. Comparison with distance-based baselines.

Method	Radius mode	Full SI-SDR (dB) ↑	Partial SI-SDRi (dB) ↑	Empty Atten. (dB) ↑
DSS [5]	Fixed	22.03	0.39	56.12
QD-TSE [7] + interval sum	Local + sum	25.99	1.12	54.10
Ours	Continuous	**29.25**	**4.40**	**62.01**

4.2. Ablation study

The ablation in Table 2 isolates the contributions of continuous radius encoding and CRC. The discrete variant uses one-hot radius codes while all other settings remain unchanged. Continuous encoding outperforms discrete encoding even without CRC, improving SI-SDR by 2.36 dB from 24.68 dB to 27.04 dB and reducing RCE from 1.04% to 0.85%. CRC improves both encoding schemes. With continuous-radius encoding, it increases SI-SDR by 2.21 dB, SI-SDRi by 1.32 dB, and attenuation by 10.88 dB, while reducing RCE to 0.46%. Combining continuous-radius encoding with CRC gives the best value for every metric. These results confirm that both continuous-radius encoding and CRC learning contribute to regional extraction performance, with their combination yielding the strongest results.

Table 2. Ablation study on radius encoding and CRC.

Radius encoding	CRC	Full SI-SDR (dB) ↑	Partial SI-SDRi (dB) ↑	Empty Atten. (dB) ↑	Cross-radius RCE (%) ↓
Discrete	×	24.68	3.04	49.68	1.04
Discrete	✓	28.75	3.15	57.70	0.81
Continuous	×	27.04	3.08	51.13	0.85
Continuous	✓	**29.25**	**4.40**	**62.01**	**0.46**

4.3. Continuous-radius analysis

Beyond the four shared comparison radii, dense 0.1-m queries examine whether outputs remain stable within unchanged target regions while preserving transitions at source boundaries (Fig. 4). Fig. 4(a) shows the normalized output change for a representative mixture. The output remains stable when the target-speaker set is unchanged and changes sharply when the radius crosses a source boundary. Fig. 4(b) shows aggregated task-oriented metrics over the test set. For each mixture with source distances $d_1 < d_2$, query positions in the empty $[0, d_1)$, partial $[d_1, d_2)$, and full $[d_2, R_{\max}]$ regions

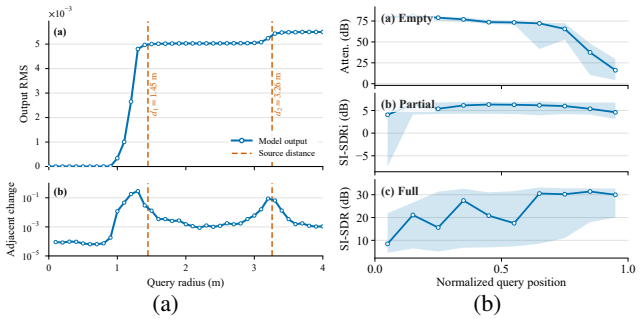


Fig. 4. Continuous-radius analysis: (a) response of a representative mixture and (b) aggregated dense-query performance.

Table 3. Evaluation under challenging conditions.

Condition	Full SI-SDR (dB) ↑	Partial SI-SDRi (dB) ↑	Empty Atten. (dB) ↑	Cross-radius RCE (%) ↓
2-spk, Clean	29.25	4.40	62.01	0.46
2-spk, 10 dB	29.05	4.23	61.76	2.31
2-spk, 5 dB	28.38	3.92	60.47	3.17
3-spk, Clean	26.17	3.19	59.28	1.12
3-spk, 10 dB	24.98	3.10	52.67	4.15

are independently normalized to $[0, 1]$ and grouped into bins. The curves and shaded areas denote the median and interquartile range, respectively. The results show strong empty-region suppression, positive partial-region SI-SDRi, and preserved full-target SI-SDR across densely sampled radii.

4.4. Evaluation under challenging conditions

Finally, Table 3 evaluates generalization to unseen three-speaker mixtures and robustness to additive noise at 5 and 10 dB SNR. When the speaker count increases from two to three, SI-SDR decreases by 3.08 dB and SI-SDRi by 1.21 dB, while attenuation decreases to 59.28 dB and RCE increases from 0.46% to 1.12%. Under additive noise at 10 dB SNR, SI-SDR decreases by 0.20 dB and SI-SDRi by 0.17 dB, while RCE increases by a factor of five from 0.46% to 2.31%. At 5 dB SNR, RCE further increases to 3.17%. Under the combined three-speaker, 10-dB condition, the model achieves 3.10 dB SI-SDRi and 52.67 dB attenuation with an RCE of 4.15%.

5. CONCLUSION

This work presents a monaural continuous-radius regional speech extraction method that directly models cumulative regional targets and supports queries within the trained range using a single model. Ablations show that continuous radius encoding outperforms one-hot radius codes, while CRC improves both extraction performance and cross-radius consistency. Dense-query analysis shows that outputs remain stable when the target-speaker set is unchanged and change sharply at source boundaries. Evaluations on measured RIRs show that the proposed method outperforms task-matched fixed-threshold and local-range baselines and remains effective with more speakers and additive noise.

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